

ting up solar plants. Three southern states, namely, Karnataka, Andhra Pradesh and the newly-formed Telangana, have allotted 500MW of solar projects each and, if all goes well, a majority of these projects will be commissioned by the end of this year or early next year.

Apart from that, several projects that were started in 2014 will be completed this year. These projects include 750MW of solar projects allotted under phase 2, batch 1 of JNNM, allotted under the state policies of Uttar Pradesh, Punjab and Madhya Pradesh. Overall, it is expected that around 2GW of solar capacity will be added in 2015.

On the way to Make in India

The 2GW growth prospect in 2015 along with the proposed five-fold increase in JNNM targets will boost solar manufacturing industry in the country. This, in turn, will help in the creation of technology hubs for manufacturing, broadly in sync with Make in India initiative.

Apart from local players, global solar manufacturers also have an eye on India as one of their preferred destinations for solar energy investment. According to industry sources, there are a host of global renewable energy firms that have shown interest in investing here. India has made it clear that it will encourage global as well as local solar manufacturers for Make in India and are offering huge incentives to these firms to start production here. Recently, we have seen a flurry of global investors investing in states like Delhi, Tamil Nadu and Kerala, among others. The numbers

Year-wise Roadmap to Achieve the Target of 100GW Solar Capacity by 2010

	2015-16	2016-17	2017-18	2018-19	2019-20	Total
Rooftop solar (GW)	3	7	9	10	11	40
Large-scale solar (GW)	4	6	8	10	12	40
Ultra-mega power projects (GW)	0	5	5	5	5	20
Total (GW)	7	18	22	25	28	100

(Source: RE-Invest by MNRE)

are expected to increase substantially in the coming months.

However, numerous challenges need to be addressed to make Make in India a reality for solar manufacturing. Off-grid solar power systems face major challenges like high capex-investment requirements, inefficient subsidy-disbursement processes and no minimum guarantee of return on investment (RoI). Moreover, it is not feasible to render any generic installation approach due to varying consumer locations and preferences. Therefore increasing solar capacity in the country requires planning and innovation to get rid of those challenges.

Bridging policy gaps

Multiple decentralised schemes are available from different ministries to facilitate solar electrification schemes. These schemes need to be integrated or linked to provide clear directives to the industry and users. A centralised system to disburse capital subsidies may further streamline the process.

Introduction of an audit system for monitoring the obligatory solar purchase by DISCOMs and heavy penalty for defaulters may ensure

adherence to the fulfillment of solar purchase obligation.

It is hardly possible to achieve the ambitious target of 100GW by floating tenders. Therefore just like the wind-energy segment, Feed in Tariff mechanism could be useful in solar as well.

The decision to drop anti-dumping duty on the import of solar cells may be a big hindrance for domestic manufacturing.

Facilitating investment

Specific solar sector schemes need to be implemented for reducing the cost of finance and increasing RoI. Sectoral cap for lending in the power sector is also applicable for solar and other renewable energy sectors. Raising finance for solar systems is a challenge with a dearth of low-interest loans due to this sectoral cap. Removal of this cap for renewable energy sector will improve leverage for the renewable industry.

Benchmark prices offered under different government subsidy schemes or tenders for solar system installation are often considerably lower than the actual cost of project implementation. This leads to disinterest among industry players about government tenders. Public-private partnerships for government-owned solar projects involving local entrepreneurs as well as electricity supply companies can be an effective model to ensure industry participation. Various forms of incentives such as tax holidays and partial risk guarantee (from the state government) can be provided to encourage solar industry.

Moreover, solar plants requiring

Unique opportunities

Smartcities. The smartcity initiative being undertaken by the government of India is likely to entail significant investments in installing renewable energy capacity.

Railways. Indian Railways plans to leverage solar energy by installing solar power systems on roofs of 500 trains as a pilot project.

Delhi solar policy draft

Delhi government in its solar policy has suggested generation of 1000MW of solar power in the capital by the year 2020.

The draft also recommends doubling this to 2GW by 2025, which is one-fifth of the state's peak power demand. It will help resolve the power crisis in the capital.